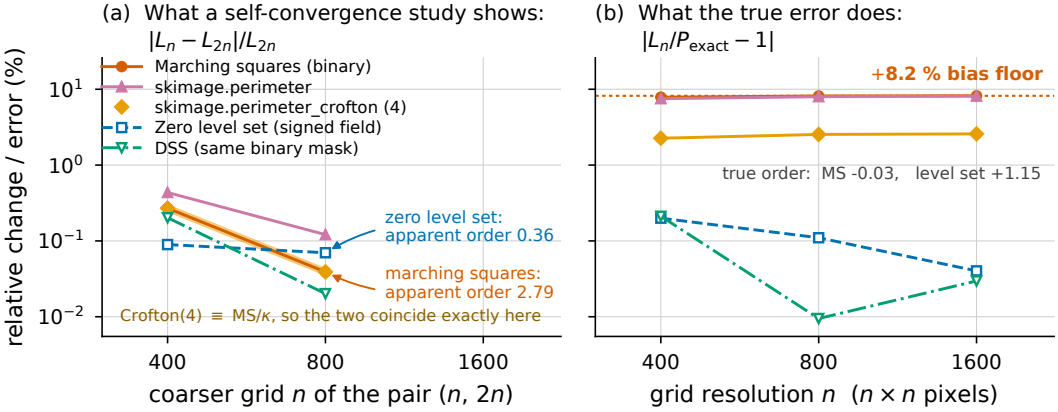
(a) What a self-convergence study shows:
$|L_n - L_{2n}|/L_{2n}$
Marching squares (binary)
skimage.perimeter
skimage.perimeter_crofton (4)
Zero level set (signed field)
DSS (same binary mask)
relative change / error (%)
10^1
10^0
10^{-1}
10^{-2}
zero level set:
apparent order 0.36
marching squares:
apparent order 2.79
Crofton(4) ≡ MS/κ, so the two coincide exactly here
400
800
1600
coarser grid n of the pair (n, $2n$)
(b) What the true error does:
$|L_n/P_{\text{exact}} - 1|$
+8.2 % bias floor
true order: MS -0.03, level set +1.15
400
800
1600
grid resolution n ($n \times n$ pixels)